Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

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Abstract

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Background

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\overline{F} = 1 - F$. Given an iid sample $X_1, \ldots, X_n$ from F, the corresponding order statistics are denoted by $X_{1:n} \le X_{2:n} \le \cdots \le X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Appendix B.1).

Definition 2.1 (Regular variation)**.** A Lebesgue measurable function $f : (0, \infty) \to \mathbb{R}$ that is eventually positive is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \mathrm{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t\to\infty} \frac{f(ty)}{f(t)} \;=\; y^\alpha.$$

The functions in RV_0 are called *slowly varying.*

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition)**.** There exists $\gamma > 0$ such that $\overline{F} \in \mathrm{RV}_{-1/\gamma}$, or equivalently the *tail quantile function*

$$U(t) \;\coloneqq\; \inf\Big\{x \in \mathbb{R} \,:\, \tfrac{1}{\overline{F}(x)} \ge t\Big\}, \qquad t > 1,$$

satisfies $U \in \mathrm{RV}_\gamma$.

Throughout, for a continuous distribution function F, we write

$$Q_X(\tau) \;\coloneqq\; F^{\leftarrow}(\tau) = \inf\{x \in \mathbb{R} : F(x) \ge \tau\}, \qquad 0 < \tau < 1.$$

With the tail quantile $U(t) = F^{\leftarrow}(1 - 1/t)$, $t > 1$, this gives

$$Q_X(\tau) = U((1-\tau)^{-1}), \qquad Q_X(1-p) = U(1/p).$$

Condition 2.2 is the Fréchet max-domain condition with extreme-value index γ (Haan and Ferreira 2006, Theorem 1.2.1); standard examples include Pareto, Burr, Student-t, Fréchet and F-distribution tails (Haan and Ferreira 2006, §2.6). For normal limits we use the standard second-order refinement.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$)**.** There exist $\gamma > 0$, a second-order parameter $\rho \leq 0$, and a measurable auxiliary function A of constant sign with $A(t) \to 0$ as $t \to \infty$, such that for every $y > 0$,

$$\lim_{t\to\infty} \frac{1}{A(t)}\left[\frac{U(ty)}{U(t)} - y^{\gamma}\right] \;=\; y^{\gamma}\,\frac{y^{\rho}-1}{\rho},$$

where the right-hand side is read as $y^{\gamma}\log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9). Since A has constant sign eventually, signed ratios $A(ty)/A(t)$ converge to y^{ρ}; the rate A is the source of the second-order bias terms below.

Condition 2.4 (Finite first-moment range)**.** In addition to Condition 2.2, the extreme-value index satisfies $\gamma \in (0, 1)$ and the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}\, X^- < \infty.$$

Under Condition 2.2, $\gamma < 1$ gives $\mathbb{E}\, X^+ < \infty$; together with the displayed lower-tail condition this is the finite-mean regime for the quantile-based expectile approximation.

2.2 Tail-index and extreme-quantile estimation

We collect the two classical estimators used throughout the thesis: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. The textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \ldots, X_n$ from F and an integer $k = k_n \in \{1, \ldots, n-1\}$, the *Hill estimator* (Hill 1975) of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) \;=\; \frac{1}{k}\sum_{i=1}^{k} \log\!\left(\frac{X_{n-i+1\,:\,n}}{X_{n-k\,:\,n}}\right). \tag{2.1}$$

Since $U(t) \to \infty$, $X_{n-k:n} > 0$ with probability tending to one for intermediate k; log-based estimators are understood on this event. For tail-index estimation one may also shift the data, but shifts must be undone for quantile or expectile estimation. The consistency result below is Haan and Ferreira (2006, Theorem 3.2.2); the normal limit is Haan and Ferreira (2006, Theorem 3.2.5) when $\rho < 0$, with the $\rho = 0$ endpoint covered by the $m = 1$ specialisation of Daouia, Padoan, et al. (2024, Theorem 1).

Theorem 2.5. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \to \infty$ and $k/n \to 0$ as $n \to \infty$. Then*

$\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. *If, in addition,* $\sqrt{k}\, A(n/k) \to \lambda \in \mathbb{R}$, *then*

$$\sqrt{k}\,(\hat{\gamma}_n(k) - \gamma) \ \xrightarrow{d}\ \mathcal{N}\Big(\frac{\lambda}{1-\rho},\ \gamma^2\Big).$$

The choice of k controls the usual bias–variance trade-off; here it is treated as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1-k/n)$ at the intermediate level $1-k/n$ — to a quantile at a very-extreme level $1-p$ with $p = p_n \to 0$ as $n \to \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \to y^\gamma$ gives the Weissman estimator

$$\widehat{q}_n^{\star}(1-p \mid k) \ = \ \Big(\frac{k}{np}\Big)^{\hat{\gamma}_n(k)} X_{n-k\,:\,n}. \tag{2.2}$$

The theorem below, the Hill-estimator specialisation of Haan and Ferreira (2006, Theorem 4.3.8), covers the far-tail regime $np = o(k)$; the case $np = O(1)$ is very-extreme in the sense that only finitely many observations are expected beyond the target.

Theorem 2.6. *Assume* $\mathcal{C}_2(\gamma, \rho, A)$ *with* $\gamma > 0$ *and* $\rho < 0$. *Let* $k = k_n \to \infty$, $k/n \to 0$, $p = p_n \to 0$ *with* $k/(np) \to \infty$ *and* $\sqrt{k}/\log(k/(np)) \to \infty$. *Suppose* $\sqrt{k}\, A(n/k) \to \lambda \in \mathbb{R}$. *Then*

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^{\star}(1-p \mid k)}{Q_X(1-p)} - 1\right) \ \xrightarrow{d}\ \mathcal{N}\Big(\frac{\lambda}{1-\rho},\ \gamma^2\Big).$$

The log-denominator is the cost of extrapolation. Up to that factor the limit is the Hill limit, which is why Daouia, Padoan, et al. (2024, §2.3) pool Weissman estimators geometrically. This pooled-Weissman law is the source result against which the rebuilt expectile estimator will be compared.

2.3 Expectiles

Expectiles were introduced by Newey and Powell (1987) as asymmetric-least-squares analogues of quantiles. Their role as coherent and elicitable risk measures is developed in Bellini et al. (2014), Ziegel (2016), Gneiting (2011), and Ehm et al. (2016).

2.3.1 Definition

Let X be a real-valued random variable with $\mathbb{E}\,|X| < \infty$, and fix a level $\tau \in (0,1)$. The τ-*expectile* of X is defined as

$$\xi_\tau(X) \ \coloneqq\ \operatorname*{arg\,min}_{\theta\in\mathbb{R}}\ \mathbb{E}[\,\eta_\tau(X-\theta) - \eta_\tau(X)\,], \qquad \eta_\tau(u) \coloneqq |\tau - \mathbb{1}\{u \le 0\}|\ u^2. \tag{2.3}$$

The subtraction of $\eta_\tau(X)$ makes the criterion finite under $\mathbb{E}\,|X| < \infty$ and does not affect the minimiser; at $\tau = 1/2$ one recovers the mean.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau\,\mathbb{E}[(X - \xi_\tau(X))_+] \;=\; (1-\tau)\,\mathbb{E}[(\xi_\tau(X) - X)_+], \tag{2.4}$$

where $u_+ = \max(u, 0)$. Thus expectiles balance upper and lower first partial moments, unlike quantiles, which balance tail probabilities.

2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

Proposition 2.7. *Let X and Y be real-valued random variables with $\mathbb{E}\,|X|, \mathbb{E}\,|Y| < \infty$. For each $\tau \in (0,1)$:*

(i) ***Existence and uniqueness.*** *$\xi_\tau(X)$ exists and is uniquely defined.*

(ii) ***Law invariance.*** *If X and Y have the same distribution, then $\xi_\tau(X) = \xi_\tau(Y)$.*

(iii) ***Mean recovery.*** *$\xi_{1/2}(X) = \mathbb{E}\,X$.*

(iv) ***Location equivariance.*** *$\xi_\tau(X + c) = \xi_\tau(X) + c$ for $c \in \mathbb{R}$.*

(v) ***Positive homogeneity.*** *$\xi_\tau(\lambda X) = \lambda\,\xi_\tau(X)$ for $\lambda \geq 0$.*

(vi) ***Sign reflection.*** *$\xi_\tau(-X) = -\xi_{1-\tau}(X)$.*

(vii) ***Monotonicity in τ.*** *The map $\tau \mapsto \xi_\tau(X)$ is nondecreasing on $(0,1)$, and is strictly increasing when X is nondegenerate, with the boundary limits $\xi_\tau(X) \to \operatorname{ess\,inf}(X)$ as $\tau \to 0$ and $\xi_\tau(X) \to \operatorname{ess\,sup}(X)$ as $\tau \to 1$ (infinite limits allowed).*

(viii) ***Monotonicity in the argument.*** *If $X \leq Y$ almost surely, then $\xi_\tau(X) \leq \xi_\tau(Y)$.*

For losses, expectiles are translation equivariant, positively homogeneous and monotone. The remaining coherence issue is subadditivity.

2.3.3 Coherence and elicitability

In the rest of the thesis X denotes a loss. The coherence literature is often written for financial positions, so in this subsection only write Y for a position and define the capital requirement

$$\rho_\tau(Y) \;=\; \xi_\tau(-Y).$$

Theorem 2.8 (Bellini et al. (2014))**.** *The functional $\rho_\tau(Y) = \xi_\tau(-Y)$ is a coherent risk measure if and only if $\tau \geq 1/2$.*

Recall that a functional is *elicitable* if it is the unique minimiser of an expected strictly consistent score (Gneiting 2011). Expectiles are elicitable on classes with finite first moment; the asymmetric squared score is strictly consistent under finite second moments, and the general scoring class has Bregman-type and Choquet-type representations (Gneiting 2011; Ehm et al. 2016). This scoring structure motivates forecast comparison, but the asymptotic theory below uses only (2.4) and the extreme expectile–quantile equivalence.

2.4 The high-expectile/high-quantile bridge

The contribution chapter uses expectile theory through the population relationship between a high expectile and the quantile of the same level. The stochastic extreme-quantile part is supplied by the Daouia–Padoan–Stupfler pooled Weissman estimator recalled in Section 2.5; the expectile-specific work is to control the deterministic bridge between ξ_τ and $Q_X(\tau)$. This section therefore records only the bridge statements needed later.

2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

Proposition 2.9 (Asymptotic equivalence; Bellini et al. (2014), Daouia, Girard, et al. (2019), Corollary 1 with $p = 2$)**.** *Suppose Condition 2.4 holds. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau\uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}. \tag{2.5}$$

The bridge requires only the finite-mean range $\gamma < 1$ and lower-tail integrability; stronger restrictions used for direct empirical expectile estimation are not part of the quantile-based bridge route. The function ψ is continuous on $(0, 1)$, with $\psi(\gamma) \to 1$ as $\gamma \downarrow 0$ and $\psi(\gamma) \to \infty$ as $\gamma \uparrow 1$. It is *not* monotone: its logarithmic derivative

$$m(\gamma) := \frac{d}{d\gamma} \log \psi(\gamma) = \frac{1}{1-\gamma} - \log\left(\gamma^{-1} - 1\right) \tag{2.6}$$

changes sign at $\gamma^\star \approx 0.2178$, where ψ reaches its minimum. In the pooled extreme-expectile theorem below, $m(\gamma)$ appears in the Taylor expansion of the bridge-Hill term; when $m(\gamma) = 0$, that term is even smaller at first order.

Under the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\mathbb{E}\, X^- < \infty$, the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia, Girard, et al. 2020, Proposition 1(i))

$$\begin{aligned}\frac{\xi_\tau(X)}{Q_X(\tau)} &= \psi(\gamma)\Big[1 + c(\gamma,\rho)\, A((1-\tau)^{-1}) + \gamma(\gamma^{-1}-1)^{\gamma}\frac{\mathbb{E}\, X}{Q_X(\tau)} \\ &\qquad + o(|A((1-\tau)^{-1})|) + o(Q_X(\tau)^{-1})\Big], \\ c(\gamma,\rho) &:= \frac{(\gamma^{-1}-1)^{-\rho}}{1-\gamma-\rho} + \frac{(\gamma^{-1}-1)^{-\rho} - 1}{\rho}.\end{aligned} \tag{2.7}$$

At $\rho = 0$ the second term in $c(\gamma, \rho)$ is read by continuity, so $c(\gamma, 0) = m(\gamma)$. The A-term is the second-order expectile–quantile bias used later to control the population bridge term. The displayed first-moment term is $O(Q_X(\tau)^{-1})$; in Chapter 3 it is controlled on the pooled-Weissman scale by the explicit bridge-rate condition.

2.4.2 Relation to earlier expectile estimators

The iid extreme-expectile literature also studies direct asymmetric-least-squares estimators, quantile-based intermediate plug-in estimators, and Weissman-type expectile extrapolations from an intermediate expectile anchor (Daouia, Girard, et al. 2018; Daouia, Girard, et al. 2020; Davison et al. 2023). Those results explain why the expectile–quantile bridge is useful, but they are not theorem inputs for the contribution chapter. The estimator analysed below performs the extreme extrapolation in the pooled-quantile component and then applies the population bridge at the target level. Accordingly, the needed expectile background is the first- and second-order bridge above, rather than a generic CLT for an intermediate expectile estimator.

2.5 The optimal pooling framework

We now recall the pooling framework of Daouia, Padoan, et al. (2024). The rebuilt thesis uses this framework as the source theory for pooled tail-index and extreme-quantile inference before applying the expectile–quantile bridge. The framework pools tail-index and extreme-quantile estimators from m samples, allowing heterogeneous sizes, second-order properties and cross-sample tail dependence. This general framework is recalled to identify the source objects and notation. The main theorem in Chapter 3 later uses only the independent common-distribution distributed special case.

2.5.1 Setup

Throughout Section 2.5, m is fixed while $n = \sum_{j=1}^{m} n_j \to \infty$. Let $\boldsymbol{X}_i = (X_{i,1}, \ldots, X_{i,m})^\top$, $i \geq 1$, be iid copies of $\boldsymbol{X} = (X_1, \ldots, X_m)^\top$. Margin j is observed for $i = 1, \ldots, n_j$ and has continuous right-heavy-tailed distribution F_j, tail quantile function $U_j = (1/\overline{F}_j)^{\leftarrow}$ and extreme-value index $\gamma_j > 0$; its order statistics are denoted $X_{1:n_j,j} \leq \cdots \leq X_{n_j:n_j,j}$. We write $Q_j(\tau) = F_j^{\leftarrow}(\tau)$, so that $U_j(t) = Q_j(1 - 1/t)$ for $t > 1$.

Tail inference in sample j uses its top $k_j + 1$ order statistics, where the effective sample size $k_j = k_j(n)$ satisfies $k_j \to \infty$ and $k_j/n_j \to 0$ as $n \to \infty$. The total effective sample size is $k = \sum_{j=1}^{m} k_j$. Assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \to b_j \in (0, \infty), \qquad \frac{k_1}{k_j} \to c_j \in (0, \infty), \tag{2.8}$$

with $b_1 = c_1 = 1$. The marginal F_j satisfies $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ (Condition 2.3). The thesis works under the *common-tail-index* assumption

$$(H_\gamma) \qquad \gamma_1 = \cdots = \gamma_m = \gamma.$$

Under this assumption, all common-γ vector limits below are expressed on the aggregate scale $\sqrt{k}$, with the constants c_j translating the local $\sqrt{k_j}$ rates.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure (Daouia, Padoan, et al. 2024, §2.1).

Condition $\mathcal{J}(\boldsymbol{R})$ (tail-copula condition). For each pair (j, ℓ) with $j \neq \ell$, there exists a function $R_{j,\ell} : [0, \infty]^2 \setminus \{(\infty, \infty)\} \to [0, \infty)$ such that, for every $(x_j, x_\ell) \in [0, \infty]^2 \setminus \{(\infty, \infty)\}$,

$$\lim_{s\to\infty} s\, \mathbb{P}\big(1 - F_j(X_j) \le x_j/s,\ 1 - F_\ell(X_\ell) \le x_\ell/s\big) \;=\; R_{j,\ell}(x_j, x_\ell).$$

The case $R_{j,\ell} \equiv 0$ is asymptotic tail independence, implied by the independent samples used later in the thesis.

2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator $\hat\gamma_j(k_j)$ as in (2.1); we collect these into the random vector

$$\hat{\boldsymbol{\gamma}}_n \;=\; \big(\hat\gamma_1(k_1), \ldots, \hat\gamma_m(k_m)\big)^\top.$$

Given a weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat\gamma_n(\boldsymbol{\omega}) \;\coloneqq\; \boldsymbol{\omega}^\top \hat{\boldsymbol{\gamma}}_n \;=\; \sum_{j=1}^m \omega_j\, \hat\gamma_j(k_j). \tag{2.9}$$

The affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$ ensures consistency for the common γ under (H_γ).

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

Theorem 2.10 (Daouia, Padoan, et al. (2024), Theorem 1)**.** *Assume $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ for every j, the tail-copula condition $\mathcal{J}(\boldsymbol{R})$, the proportionality conditions (2.8), and $\sqrt{k_j}\, A_j(n_j/k_j) \to \lambda_j \in \mathbb{R}$ for every j. Then*

$$\big(\sqrt{k_1}(\hat\gamma_1(k_1) - \gamma_1), \ldots, \sqrt{k_m}(\hat\gamma_m(k_m) - \gamma_m)\big)^\top \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{B}, \boldsymbol{V}),$$

with $\boldsymbol{B} = \big(\lambda_j/(1-\rho_j)\big)_{j=1}^m$ and $\boldsymbol{V}$ the symmetric matrix whose entries are

$$V_{j,\ell} \;=\; \begin{cases} \gamma_j^2, & j = \ell, \\ \dfrac{\gamma_j \gamma_\ell}{\max(b_j, b_\ell)\sqrt{c_j c_\ell}}\, R_{j,\ell}(b_j c_\ell,\, b_\ell c_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so $V_{\ell,j} = V_{j,\ell}$ for $\ell > j$. Suppose, in addition, the common-γ condition (H_γ). Then the same marginal estimators, now viewed on the common $\sqrt{k}$ scale, satisfy

$$\sqrt{k}\,(\hat{\boldsymbol{\gamma}}_n - \gamma \mathbf{1}) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{B}_{\boldsymbol{c}}, \boldsymbol{V}_{\boldsymbol{c}}),$$

where

$$(\boldsymbol{B_c})_j = \sqrt{c_j \sum_{i=1}^{m} c_i^{-1} \frac{\lambda_j}{1-\rho_j}} \qquad j = 1, \dots, m.$$

The covariance matrix $\boldsymbol{V_c}$ *is given by*

$$(\boldsymbol{V_c})_{j,\ell} = \left(\sum_{i=1}^{m} c_i^{-1}\right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \dfrac{\gamma^2}{\max(b_j, b_\ell)} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell, \end{cases}$$

with $(\boldsymbol{V_c})_{\ell,j} = (\boldsymbol{V_c})_{j,\ell}$ *for* $\ell > j$*. Consequently, for any* $\boldsymbol{\omega}$ *with* $\boldsymbol{\omega}^\top \mathbf{1} = 1$*,*

$$\sqrt{k}\,(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c},\ \boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}).$$

For independent samples, all off-diagonal entries vanish and $\boldsymbol{V_c}$ becomes the diagonal matrix

$$\boldsymbol{V_c} \ = \ \gamma^2 \left(\sum_{i=1}^{m} c_i^{-1}\right) \operatorname{diag}(c_1, \dots, c_m). \tag{2.10}$$

Equation (2.10) is the covariance form to be used when the rebuilt main theorem is specialised to independent samples.

2.5.3 Variance- and AMSE-optimal weights

The relevant weights solve quadratic programmes under the affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$; unless stated otherwise, they need not be nonnegative. In Chapter 3, these formulas are applied only to the pooled-Weissman weights; they do not define or optimise the separate bridge-Hill weights introduced there.

Proposition 2.11 (Optimal weights; Daouia, Padoan, et al. (2024), Theorem 1; estimated-weight implementation in §2.2 and Corollary 1)**.** *Suppose* $\boldsymbol{V_c}$ *is positive definite. Under the common-*γ *condition* (H_γ)*:*

(i) The variance-optimal weights, minimising $\boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}$ *subject to* $\boldsymbol{\omega}^\top \mathbf{1} = 1$*, are*

$$\boldsymbol{\omega}^{\mathrm{Var}} \ = \ \frac{\boldsymbol{V_c}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{V_c}^{-1} \mathbf{1}},$$

with corresponding asymptotic variance $\boldsymbol{\omega}^{\mathrm{Var},\top} \boldsymbol{V_c} \boldsymbol{\omega}^{\mathrm{Var}} = (\mathbf{1}^\top \boldsymbol{V_c}^{-1} \mathbf{1})^{-1}$*.*

(ii) The AMSE-optimal weights, minimising $(\boldsymbol{\omega}^\top \boldsymbol{B_c})^2 + \boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}$ *subject to* $\boldsymbol{\omega}^\top \mathbf{1} = 1$*, are*

$$\boldsymbol{\omega}^{\mathrm{AMSE}} \ = \ \frac{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1} \boldsymbol{B_c})\, \boldsymbol{V_c}^{-1}\mathbf{1} - (\mathbf{1}^\top \boldsymbol{V_c}^{-1} \boldsymbol{B_c})\, \boldsymbol{V_c}^{-1} \boldsymbol{B_c}}{(1 + \boldsymbol{B_c}^\top \boldsymbol{V_c}^{-1} \boldsymbol{B_c})(\mathbf{1}^\top \boldsymbol{V_c}^{-1} \mathbf{1}) - (\mathbf{1}^\top \boldsymbol{V_c}^{-1} \boldsymbol{B_c})^2}.$$

In the independent-samples case, $\boldsymbol{V_c}$ is diagonal (Equation (2.10)) and the variance-optimal weights reduce to

$$\omega_j^{\mathrm{Var}} \;=\; \left(\sum_{i=1}^{m} c_i^{-1}\right)^{-1} c_j^{-1} \;\sim\; \frac{k_j}{k}, \tag{2.11}$$

where the equivalence uses $c_i^{-1} = \lim k_i/k_1$. The finite-sample distributed version is

$$\tilde{\boldsymbol{\omega}}_n^{\mathrm{Var}} = (k_1/k, \ldots, k_m/k)^\top,$$

which is exactly normalised and converges to $\boldsymbol{\omega}^{\mathrm{Var}}$.

Feasible weights replace $\boldsymbol{V_c}$ and $\boldsymbol{B_c}$ by consistent plug-ins (Daouia, Padoan, et al. 2024, §2.2). If $\hat{\boldsymbol{\omega}}^\top \mathbf{1} = 1$ exactly and $\hat{\boldsymbol{\omega}} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$, then $\hat{\gamma}_n(\hat{\boldsymbol{\omega}})$ is $\sqrt{k}$-equivalent to $\hat{\gamma}_n(\boldsymbol{\omega})$; the difficult practical step is stable estimation of $\boldsymbol{V_c}$ and the bias quantities.

2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile $q(1-p)$ with $p = p(n) \to 0$, each sample contributes a Weissman estimator

$$\hat{q}_j^\star(1-p \mid k_j) \;=\; \left(\frac{k_j}{n_j\, p}\right)^{\hat{\gamma}_j(k_j)} X_{n_j-k_j:n_j,\,j}.$$

Because Weissman extrapolation is multiplicative, pooling is geometric on the original scale, equivalently linear on the log scale. It requires asymptotic equivalence of the marginal extreme quantiles:

Condition 2.12 (Tail homoskedasticity ($\mathcal{H}$))**.** For every $j \neq \ell$, $U_j(t)/U_\ell(t) \to 1$ as $t \to \infty$.

Equivalently, $Q_j(1-p)/Q_\ell(1-p) \to 1$ as $p \downarrow 0$. The pooled extreme-quantile estimator of Daouia, Padoan, et al. (2024, §2.3) is

$$\widehat{q}_n^\star(1-p \mid \boldsymbol{\omega}) \;:=\; \prod_{j=1}^{m} \left[\hat{q}_j^\star(1-p \mid k_j)\right]^{\omega_j}. \tag{2.12}$$

Taking logs reduces pooling to weighted sums of the local identities $\log \hat{q}_j^\star(1-p \mid k_j) = \log(k_j/(n_j p))\, \hat{\gamma}_j(k_j) + \log X_{n_j-k_j:n_j,\,j}$; so the Hill part is handled by Theorem 2.10; (2.8) gives $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$.

Theorem 2.13 (Daouia, Padoan, et al. (2024), Theorem 2)**.** *Assume the hypotheses of Theorem 2.10, the common-γ condition (H_γ), the tail-homoskedasticity condition $(\mathcal{H})$, and, in addition, $\rho_j < 0$ for every $j = 1, \ldots, m$. Let $p = p(n) \to 0$ with $k/(np) \to \infty$ and $\sqrt{k}/\log(k/(np)) \to \infty$. Then for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$ and any $j \in \{1, \ldots, m\}$,*

$$\frac{\sqrt{k}}{\log(k/(np))} \left(\frac{\widehat{q}_n^\star(1-p \mid \boldsymbol{\omega})}{Q_j(1-p)} - 1 \right) \;\xrightarrow{d}\; \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c},\; \boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}).$$

The strict $\rho_j < 0$ assumption is part of the Weissman theorem. Together with the bias-rate assumptions $\sqrt{k_j}\, A_j(n_j/k_j) \to \lambda_j$ in Theorem 2.10, it also supplies the rate needed to recenter by any marginal $Q_j(1-p)$ under $(\mathcal{H})$ (Daouia, Padoan, et al. 2024, Supplement, Lemma B.2):

for any j, ℓ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(\,|A_j(1/p)|\,) + O(\,|A_\ell(1/p)|\,) = o(k^{-1/2}),$$

along the Weissman sequence.

Thus the pooled Weissman and pooled Hill limits share the same mean, variance and optimal weights up to the log-rate factor. This shared structure is the reason the contribution chapter can import the Daouia–Padoan–Stupfler weight criteria for the pooled-Weissman component.

2.5.5 Estimated weights and plug-in objects

The optimal weights above are oracle quantities because they depend on the asymptotic bias and covariance objects. For inference they are replaced either by deterministic allocation weights, such as $(k_1/k, \ldots, k_m/k)^\top$ in the independent distributed case, or by estimated weights based on plug-in estimates of the same objects. The following result records the stability property used in the contribution chapter. There, estimated weights are used only in the pooled-Weissman role; the bridge-Hill vector remains deterministic.

Proposition 2.14 (Estimated pooling weights; Daouia, Padoan, et al. (2024), Supplement, Theorem A.1; Theorem 2 and Corollary 8)**.** *Assume the common-γ conditions of Theorem 2.10. Let $\widehat{\boldsymbol{\omega}}_n$ be a random weight vector satisfying*

$$\widehat{\boldsymbol{\omega}}_n^\top \mathbf{1} = 1, \qquad \widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega},$$

for some deterministic $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$. Then

$$\sqrt{k}\,(\hat{\gamma}_n(\widehat{\boldsymbol{\omega}}_n) - \gamma) \ \xrightarrow{d}\ \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c}, \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}).$$

If, in addition, the assumptions of Theorem 2.13 hold, then

$$\frac{\sqrt{k}}{\log(k/(np))}\left(\frac{\widehat{q}_n^\star(1-p \mid \widehat{\boldsymbol{\omega}}_n)}{Q_j(1-p)} - 1\right) \ \xrightarrow{d}\ \mathcal{N}(\boldsymbol{\omega}^\top \boldsymbol{B_c}, \boldsymbol{\omega}^\top \boldsymbol{V_c}\boldsymbol{\omega}), \qquad j = 1, \ldots, m.$$

In the independent common-distribution distributed case, the same statement holds with the diagonal covariance matrix (2.10) *and the distributed bias constants of Daouia, Padoan, et al. (2024, Corollary 5).*

The supplement's Theorem A.1 is the generic estimated-weight transfer: exact affine normalisation and convergence in probability of the weights make the random-weight pooled estimator asymptotically equivalent to the pooled estimator with limiting deterministic weights. Applied to Theorem 2.10, this gives the Hill statement. The Weissman statement is the corresponding random-weight version of Theorem 2.13; in the distributed common-marginal case it is the published Corollary 8 of Daouia, Padoan, et al. (2024).

The next proposition separates the probabilistic consistency of the estimated bias and variance objects from the algebra of applying a chosen weight vector.

Proposition 2.15 (Plug-in bias and variance)**.** *Let $\widehat{\boldsymbol{B}}_{\boldsymbol{c}n}$ and $\widehat{\boldsymbol{V}}_{\boldsymbol{c}n}$ be estimators of $\boldsymbol{B_c}$ and $\boldsymbol{V_c}$ satisfying*

$$\widehat{\boldsymbol{B}}_{\boldsymbol{c}n} \xrightarrow{\mathbb{P}} \boldsymbol{B_c}, \qquad \widehat{\boldsymbol{V}}_{\boldsymbol{c}n} \xrightarrow{\mathbb{P}} \boldsymbol{V_c},$$

and let $\widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. Define

$$\widehat{B}_{\widehat{\boldsymbol{\omega}},n} = \widehat{\boldsymbol{\omega}}_n^\top \widehat{\boldsymbol{B}}_{\boldsymbol{c}n}, \qquad \widehat{V}_{\widehat{\boldsymbol{\omega}},n} = \widehat{\boldsymbol{\omega}}_n^\top \widehat{\boldsymbol{V}}_{\boldsymbol{c}n} \widehat{\boldsymbol{\omega}}_n.$$

Then

$$\widehat{B}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}^\top \boldsymbol{B_c}, \qquad \widehat{V}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} \boldsymbol{\omega}^\top \boldsymbol{V_c} \boldsymbol{\omega}.$$

All vectors and matrices have fixed dimension. The maps $(w, b) \mapsto w^\top b$ and $(w, V) \mapsto w^\top V w$ are continuous, so the result follows from the continuous mapping theorem.

Daouia, Padoan, et al. (2024, §2.2 and Corollary 1) give one source route to the consistency assumptions in Proposition 2.15 through estimated second-order and tail-dependence quantities. That route is not assumption-free: in the source it is stated under $\rho_j < 0$, the parametric second-order form $A_j(t) = \gamma \beta_j t^{\rho_j}$, positive definiteness of the asymptotic covariance matrix, consistency of the β_j estimators, and $(\hat{\rho}_j - \rho_j) \log n_j = o_P(1)$. In the common-distribution distributed case, their Corollaries 6–7 give the corresponding deterministic variance allocation and plug-in AMSE constructions under the analogous common second-order parameter consistency assumptions.

2.5.6 Source diagnostics and intervals

Daouia, Padoan, et al. (2024) also develop tail-homogeneity tests for the common-γ condition, confidence intervals for the common tail index, and log-scale intervals for pooled Weissman quantiles. These tools are useful diagnostics in the source pooling framework. The expectile interval in Chapter 3, however, is not imported as a direct restatement of a DPS quantile interval; it is obtained by Slutsky's theorem and inversion after the exact expectile decomposition.

2.5.7 Distributed inference as a special case

In the independent common-distribution distributed setting, the samples are iid subdivisions of one common population. Then:

- the common-γ condition (H_γ) and the tail-homoskedasticity condition $(\mathcal{H})$ are trivially satisfied;
- the tail copula $R_{j,\ell}$ between any pair of distinct samples vanishes identically (independence implies tail independence);
- $\boldsymbol{V_c}$ in Theorem 2.10 becomes the diagonal matrix (2.10);
- the variance-optimal weights of Proposition 2.11 reduce to $\omega_j^{\mathrm{Var}} \propto k_j$ as in (2.11).

Under the stronger balanced-allocation assumptions studied by Daouia, Padoan, et al. (2024, Corollary 6, Theorem 3 and Corollary 8), the variance-optimally pooled Hill and pooled Weissman estimators match the corresponding unfeasible estimators built from the entire combined sample at first order. This oracle equivalence is useful benchmark context for distributed inference. The main theorem in Chapter 3 uses the more direct consequence needed for the expectile problem: in the independent common-distribution case, the pooled-Weissman bias and variance objects are the diagonal distributed objects displayed above.

Chapter 3

Pooled extreme expectiles

This chapter develops the pooled extreme-expectile estimator studied in the thesis. We work in the finite-m distributed iid setting of Section 2.5.7: each machine contains observations from the same continuous heavy-tailed loss distribution, and the target is the common marginal expectile ξ_{τ_n}. In this setting the population target is unambiguous, and the pooled extreme-quantile component can be handled through the distributed pooled-Weissman theorem of Daouia, Padoan, et al. (2024, Corollary 8).

The construction combines the geometrically pooled Weissman estimator with the asymptotic bridge between high expectiles and high quantiles. The role of the chapter is to show, from an exact log decomposition, which terms remain visible on the pooled-Weissman scale. Under the bridge-rate condition imposed below, the first-order law is inherited from the pooled extreme-quantile component. Accordingly, first-order weight optimisation concerns the Daouia–Padoan–Stupfler weights in the pooled-Weissman part.

3.1 Estimator and exact decomposition

Let $p_n = 1 - \tau_n$, $n = \sum_{j=1}^m n_j$, $k = \sum_{j=1}^m k_j$, and

$$\ell_n = \log\left(\frac{k}{np_n}\right).$$

For each machine j, let $\hat{\gamma}_j(k_j)$ be the local Hill estimator and

$$\hat{q}_j^\star(1 - p_n \mid k_j) = \left(\frac{k_j}{n_j p_n}\right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}$$

be the local Weissman estimator. For an affine weight vector $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the geometrically pooled Weissman estimator is

$$\widehat{q}_n^\star(1 - p_n \mid \boldsymbol{\omega}) = \prod_{j=1}^m \left[\hat{q}_j^\star(1 - p_n \mid k_j)\right]^{\omega_j}.$$

For another affine weight vector $\boldsymbol{\nu}^\top \mathbf{1} = 1$, define the pooled Hill estimator used in the outer bridge by

$$\hat{\gamma}_n(\boldsymbol{\nu}) = \sum_{j=1}^{m} \nu_j \hat{\gamma}_j(k_j).$$

The two-weight pooled extreme-expectile estimator is

$$\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}, \boldsymbol{\omega}) = \psi(\hat{\gamma}_n(\boldsymbol{\nu}))\, \widehat{q}^{\star}_n(1 - p_n \mid \boldsymbol{\omega}), \qquad \psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}. \tag{3.1}$$

The vector $\boldsymbol{\omega}$ weights the pooled-Weissman quantile estimator. The vector $\boldsymbol{\nu}$ weights only the Hill estimator appearing in the outer expectile bridge. The diagonal construction $\boldsymbol{\nu} = \boldsymbol{\omega}$ is a special case, not a first-order necessity.

Lemma 3.1 (Exact log decomposition)**.** *Let $Q(1 - p_n)$ denote the common marginal quantile and let ξ_{τ_n} denote the corresponding expectile. Then*

$$\log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}, \boldsymbol{\omega})}{\xi_{\tau_n}} = A_n(\boldsymbol{\omega}) + B_n(\boldsymbol{\nu}) - C_n, \tag{3.2}$$

where

$$A_n(\boldsymbol{\omega}) = \log \frac{\widehat{q}^{\star}_n(1 - p_n \mid \boldsymbol{\omega})}{Q(1 - p_n)}, \qquad B_n(\boldsymbol{\nu}) = \log \psi(\hat{\gamma}_n(\boldsymbol{\nu})) - \log \psi(\gamma),$$

and

$$C_n = \log \frac{\xi_{\tau_n}}{\psi(\gamma) Q(1 - p_n)}.$$

Proof. By the definition of the estimator,

$$\begin{aligned} \log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}, \boldsymbol{\omega})}{\xi_{\tau_n}} &= \log \frac{\psi(\hat{\gamma}_n(\boldsymbol{\nu}))\widehat{q}^{\star}_n(1 - p_n \mid \boldsymbol{\omega})}{\xi_{\tau_n}} \\ &= \log \psi(\hat{\gamma}_n(\boldsymbol{\nu})) + \log \widehat{q}^{\star}_n(1 - p_n \mid \boldsymbol{\omega}) - \log \xi_{\tau_n}. \end{aligned}$$

Insert and subtract the two deterministic quantities $\log Q(1 - p_n)$ and $\log \psi(\gamma)$. This gives

$$\begin{aligned} \log \frac{\widehat{\xi}^{\text{pool},\star}_{\tau_n}(\boldsymbol{\nu}, \boldsymbol{\omega})}{\xi_{\tau_n}} &= \{\log \widehat{q}^{\star}_n(1 - p_n \mid \boldsymbol{\omega}) - \log Q(1 - p_n)\} \\ &\quad + \{\log \psi(\hat{\gamma}_n(\boldsymbol{\nu})) - \log \psi(\gamma)\} \\ &\quad + \{\log \psi(\gamma) + \log Q(1 - p_n) - \log \xi_{\tau_n}\}. \end{aligned}$$

The first bracket is $A_n(\boldsymbol{\omega})$, the second bracket is $B_n(\boldsymbol{\nu})$, and the third bracket equals

$$\log \frac{\psi(\gamma) Q(1 - p_n)}{\xi_{\tau_n}} = -\log \frac{\xi_{\tau_n}}{\psi(\gamma) Q(1 - p_n)} = -C_n.$$

Combining the three brackets yields (3.2). □

3.2 A deterministic two-weight theorem

The following theorem gives the first-order limit under a rate condition that makes the deterministic population gap between expectiles and quantiles negligible on the pooled-Weissman scale.

Theorem 3.2 (Pooled extreme expectile, deterministic weights)**.** *Let m be fixed. Suppose the observations across and within machines are iid copies of a common continuous loss variable X. Let $U(t) = Q(1 - 1/t)$ be the common tail quantile, and assume $\mathcal{C}_2(\gamma, \rho, A)$ with $0 < \gamma < 1$ and $\rho < 0$. Assume $\mathbb{E}\, X^- < \infty$ and $U(t) > 0$ for all sufficiently large t.*

Let $n = \sum_{j=1}^m n_j$, $k = \sum_{j=1}^m k_j$, and assume

$$\frac{n_1}{n_j} \to b_j \in (0, \infty), \qquad \frac{k_1}{k_j} \to c_j \in (0, \infty),$$

with $b_1 = c_1 = 1$. Suppose

$$k \to \infty, \qquad \frac{k}{n} \to 0, \qquad \sqrt{k}\, A(n/k) \to \lambda \in \mathbb{R}.$$

Let $p_n = 1 - \tau_n$, $\ell_n = \log\{k/(np_n)\}$, and assume the extreme Weissman regime

$$p_n \to 0, \qquad \frac{k}{np_n} \to \infty, \qquad \frac{\sqrt{k}}{\ell_n} \to \infty.$$

Finally impose the expectile-bridge rate condition

$$\eta_n = \frac{\sqrt{k}}{\ell_n Q(1 - p_n)} \to 0. \tag{3.3}$$

Let $\boldsymbol{\nu}_n, \boldsymbol{\omega}_n \in \mathbb{R}^m$ be deterministic weight sequences satisfying exact affine normalisation and finite-dimensional convergence,

$$\boldsymbol{\nu}_n^\top \mathbf{1} = 1, \qquad \boldsymbol{\omega}_n^\top \mathbf{1} = 1, \qquad \boldsymbol{\nu}_n \to \boldsymbol{\nu}, \qquad \boldsymbol{\omega}_n \to \boldsymbol{\omega},$$

for finite vectors $\boldsymbol{\nu}, \boldsymbol{\omega} \in \mathbb{R}^m$. Fixed deterministic weights are included by taking constant sequences. No non-negativity is imposed unless stated separately. Then the estimator (3.1) satisfies

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\widehat{\xi}^{\mathrm{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}), \tag{3.4}$$

where

$$B_{\boldsymbol{\omega}} = \frac{\lambda}{1-\rho} \sum_{j=1}^m d_j^\rho \omega_j, \qquad V_{\boldsymbol{\omega}} = \gamma^2 \left(\sum_{j=1}^m c_j^{-1} \right) \left(\sum_{j=1}^m c_j \omega_j^2 \right),$$

and

$$d_j = \frac{c_j}{b_j} \frac{\sum_{i=1}^m c_i^{-1}}{\sum_{i=1}^m b_i^{-1}}.$$

Moreover,

$$\frac{\sqrt{k}}{\ell_n}\left(\frac{\widehat{\xi}^{\mathrm{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n,\boldsymbol{\omega}_n)}{\xi_{\tau_n}}-1\right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}},V_{\boldsymbol{\omega}}). \tag{3.5}$$

Proof. Start from the exact identity (3.2). We treat the three terms separately on the scale $s_n=\sqrt{k}/\ell_n$.

For $A_n(\boldsymbol{\omega}_n)$, use the random-weight version of the common-marginal pooled-Weissman theorem recalled in Proposition 2.14, applied to the exactly normalised deterministic sequence $\boldsymbol{\omega}_n\to\boldsymbol{\omega}$. It gives

$$\frac{\sqrt{k}}{\ell_n}\left(\frac{\widehat{q}^{\star}_n(1-p_n\mid\boldsymbol{\omega}_n)}{Q(1-p_n)}-1\right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}},V_{\boldsymbol{\omega}}).$$

Set

$$x_n=\frac{\widehat{q}^{\star}_n(1-p_n\mid\boldsymbol{\omega}_n)}{Q(1-p_n)}-1.$$

The condition $\sqrt{k}/\ell_n\to\infty$ implies $x_n=O_P(\ell_n/\sqrt{k})=o_P(1)$. Hence

$$\log(1+x_n)=x_n+O_P(x_n^2)$$

and

$$\frac{\sqrt{k}}{\ell_n}\{\log(1+x_n)-x_n\}=O_P\left(\frac{\ell_n}{\sqrt{k}}\right)=o_P(1).$$

Therefore

$$\frac{\sqrt{k}}{\ell_n}A_n(\boldsymbol{\omega}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}},V_{\boldsymbol{\omega}}). \tag{3.6}$$

For $B_n(\boldsymbol{\nu}_n)$, write

$$g(x)=\log\psi(x)=-x\log(x^{-1}-1).$$

On every compact subinterval of $(0,1)$, g is twice continuously differentiable, with

$$g'(x)=m(x)=\frac{1}{1-x}-\log(x^{-1}-1),\qquad g''(x)=\frac{1}{(1-x)^2}+\frac{1}{1-x}+\frac{1}{x}.$$

By the vector pooled Hill CLT in Theorem 2.10,

$$\sqrt{k}\{\hat{\boldsymbol{\gamma}}_n-\gamma\mathbf{1}\}=O_P(1).$$

Since $\boldsymbol{\nu}_n\to\boldsymbol{\nu}$, the sequence $(\boldsymbol{\nu}_n)$ is bounded, and

$$\sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\nu}_n)-\gamma\}=\boldsymbol{\nu}_n^{\top}\sqrt{k}\{\hat{\boldsymbol{\gamma}}_n-\gamma\mathbf{1}\}=O_P(1).$$

Thus $\hat{\gamma}_n(\boldsymbol{\nu}_n)\xrightarrow{\mathbb{P}}\gamma$. Hence, with probability tending to one, the line segment between $\hat{\gamma}_n(\boldsymbol{\nu}_n)$ and γ lies in a fixed compact subset of $(0,1)$. Taylor's theorem on that event gives

$$B_n(\boldsymbol{\nu}_n)=m(\gamma)\{\hat{\gamma}_n(\boldsymbol{\nu}_n)-\gamma\}+O_P\Big(\{\hat{\gamma}_n(\boldsymbol{\nu}_n)-\gamma\}^2\Big)=m(\gamma)\{\hat{\gamma}_n(\boldsymbol{\nu}_n)-\gamma\}+O_P(k^{-1}).$$

Consequently

$$\frac{\sqrt{k}}{\ell_n} B_n(\boldsymbol{\nu}_n) = \frac{m(\gamma)}{\ell_n} \sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\nu}_n) - \gamma\} + O_P\left(\frac{1}{\ell_n \sqrt{k}}\right) = o_P(1), \tag{3.7}$$

because $\ell_n \to \infty$, which follows from $k/(np_n) \to \infty$. If $m(\gamma) = 0$, the displayed conclusion follows from the quadratic remainder alone.

It remains to control C_n. Put

$$a_n = A(1/p_n), \qquad b_n = Q(1-p_n)^{-1}, \qquad \alpha_\gamma = \gamma(\gamma^{-1} - 1)^\gamma, \qquad \mu = \mathbb{E}\, X.$$

The second-order expectile–quantile expansion (2.7) gives, after dividing by $\psi(\gamma)$,

$$\Delta_n := \frac{\xi_{\tau_n}}{\psi(\gamma) Q(1-p_n)} - 1 = c(\gamma, \rho) a_n + \alpha_\gamma \mu b_n + o(|a_n|) + o(b_n).$$

Since $\Delta_n \to 0$,

$$C_n = \log(1 + \Delta_n) = \Delta_n + O(\Delta_n^2).$$

The regular variation of A, the strict inequality $\rho < 0$, and

$$\frac{1/p_n}{n/k} = \frac{k}{np_n} \to \infty$$

imply

$$\frac{A(1/p_n)}{A(n/k)} = \left(\frac{k}{np_n}\right)^\rho \{1 + o(1)\} \to 0.$$

Together with $\sqrt{k} A(n/k) \to \lambda$, this yields

$$\frac{\sqrt{k}}{\ell_n} a_n \to 0.$$

The imposed bridge-rate condition (3.3) is exactly

$$\frac{\sqrt{k}}{\ell_n} b_n \to 0.$$

Therefore all linear terms in C_n are $o(1)$ after multiplication by $\sqrt{k}/\ell_n$. The logarithmic square remainder is also negligible, since

$$\frac{\sqrt{k}}{\ell_n} (|a_n| + b_n)^2 \to 0$$

follows from $(\sqrt{k}/\ell_n)|a_n| \to 0$, $(\sqrt{k}/\ell_n) b_n \to 0$, and $|a_n| + b_n \to 0$. Hence

$$\frac{\sqrt{k}}{\ell_n} C_n \to 0. \tag{3.8}$$

Combining (3.6), (3.7), and (3.8) in the exact decomposition gives (3.4) by Slutsky's theorem.

For the relative-error statement, define

$$L_n = \log \frac{\widehat{\xi}^{\mathrm{pool},\star}_{\tau_n}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}}.$$

The log limit implies $L_n = O_P(\ell_n/\sqrt{k}) = o_P(1)$. Therefore

$$e^{L_n} - 1 = L_n + O_P(L_n^2)$$

and

$$\frac{\sqrt{k}}{\ell_n}\{e^{L_n} - 1 - L_n\} = O_P\left(\frac{\ell_n}{\sqrt{k}}\right) = o_P(1).$$

This proves (3.5). □

Remark 3.3 (What the theorem does and does not identify). The limit in Theorem 3.2 depends on $\boldsymbol{\omega}$, the limiting geometric pooled-Weissman weights, but not on $\boldsymbol{\nu}$, the limiting bridge-Hill weights. Thus $\boldsymbol{\nu}$ is first-order unidentified in the present regime. The diagonal estimator $\boldsymbol{\nu}_n = \boldsymbol{\omega}_n$ is a permissible deterministic convention when the common deterministic sequence satisfies the theorem's convergence condition, and the practical convention used for plug-in inference below is

$$\boldsymbol{\nu}_n^k = \left(\frac{k_1}{k}, \ldots, \frac{k_m}{k}\right)^\top.$$

Neither convention is an optimality result for $\boldsymbol{\nu}$. Any optimal choice of bridge-Hill weights would require a lower-order theory not developed here.

3.3 Weight consequences

The preceding theorem determines the relevant first-order weight criterion. The leading bias and variance coincide with those of the Daouia–Padoan–Stupfler pooled-Weissman component in the distributed common-marginal case. Define

$$\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}} = \frac{\lambda}{1-\rho}(d_1^\rho, \ldots, d_m^\rho)^\top, \qquad \boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}} = \gamma^2 \left(\sum_{i=1}^m c_i^{-1}\right) \mathrm{diag}(c_1, \ldots, c_m).$$

Then $B_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top \boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}}$ and $V_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}} \boldsymbol{\omega}$.

Corollary 3.4 (First-order criteria for $\boldsymbol{\omega}$)**.** *Under the assumptions of Theorem 3.2, the first-order variance criterion for the pooled extreme-expectile estimator is*

$$\boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}} \boldsymbol{\omega},$$

and the first-order AMSE criterion is

$$(\boldsymbol{\omega}^\top \boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})^2 + \boldsymbol{\omega}^\top \boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}} \boldsymbol{\omega},$$

both under the affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$. Hence the deterministic/oracle variance- and AMSE-

optimal formulas of Daouia, Padoan, et al. (2024) apply to $\boldsymbol{\omega}$. These criteria concern $\boldsymbol{\omega}$ only; the bridge-Hill weights $\boldsymbol{\nu}$ do not enter because $B_n(\boldsymbol{\nu}_n)$ is lower order under (3.3). In particular,

$$\boldsymbol{\omega}^{\mathrm{Var}} = \frac{(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\mathbf{1}}{\mathbf{1}^\top(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\mathbf{1}},$$

so, in the independent distributed case,

$$\omega_j^{\mathrm{Var}} = \frac{c_j^{-1}}{\sum_{i=1}^m c_i^{-1}}, \qquad j = 1, \dots, m.$$

If $\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}}$ is positive definite, the AMSE-optimal weights are

$$\boldsymbol{\omega}^{\mathrm{AMSE}} = \frac{(1 + (\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})^\top(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\mathbf{1} - (\mathbf{1}^\top(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}}}{(1 + (\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})^\top(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})(\mathbf{1}^\top(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\mathbf{1}) - (\mathbf{1}^\top(\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}})^{-1}\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}})^2}.$$

No first-order variance or AMSE criterion for $\boldsymbol{\nu}$ is identified by Theorem 3.2.

Proof. By Theorem 3.2,

$$\frac{\sqrt{k}}{\ell_n}\log\frac{\widehat{\xi}_{\tau_n}^{\mathrm{pool},\star}(\boldsymbol{\nu}_n, \boldsymbol{\omega}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Thus the scaled first-order mean-square criterion is $B_{\boldsymbol{\omega}}^2 + V_{\boldsymbol{\omega}}$, and the unscaled criterion differs only by the common factor ℓ_n^2/k. The same criterion applies to relative error by (3.5). Substituting $B_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top\boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}}$ and $V_{\boldsymbol{\omega}} = \boldsymbol{\omega}^\top\boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}}\boldsymbol{\omega}$ gives exactly the quadratic criteria recalled in Proposition 2.11. The variance and AMSE formulas follow from that result. □

3.4 Estimated pooled-Weissman weights and inference

The deterministic theorem is the base result. Estimated weights are introduced only in the pooled-Weissman role $\boldsymbol{\omega}$, where Daouia, Padoan, et al. (2024) already provide random-weight quantile limits under their own consistency conditions. The bridge-Hill vector remains deterministic in this chapter. When the estimated weights or plug-in bias and variance objects are taken from the Daouia–Padoan–Stupfler constructions, the source-side consistency assumptions recalled in Section 2.5.5 are imposed.

The only new source input is the random-weight version of the Daouia–Padoan–Stupfler pooled-Weissman limit: exact affine normalisation $\widehat{\boldsymbol{\omega}}_n^\top\mathbf{1} = 1$ and convergence $\widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$ give the same first-order quantile limit as the deterministic vector $\boldsymbol{\omega}$. The rest of the argument is a transfer through the exact decomposition.

Corollary 3.5 (Estimated-$\boldsymbol{\omega}$ transfer)**.** *Assume the conditions of Theorem 3.2, with the deterministic pooled-Weissman weight sequence in the estimator replaced by a random $\widehat{\boldsymbol{\omega}}_n$ satisfying the estimated-weight conditions of Proposition 2.14, with limiting vector $\boldsymbol{\omega}$. Let $\boldsymbol{\nu}_n$ be any deterministic bridge-Hill weight sequence satisfying the exact affine normalisation and convergence* *conditions of Theorem 3.2; in particular one may take* $\boldsymbol{\nu}_n = \boldsymbol{\nu}_n^k = (k_1/k, \dots, k_m/k)^\top$. *Then*

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\widehat{\xi}_{\tau_n}^{\mathrm{pool},\star}(\boldsymbol{\nu}_n, \widehat{\boldsymbol{\omega}}_n)}{\xi_{\tau_n}} \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

The same limit holds on relative-error scale.

Proof. The exact decomposition becomes

$$\log \frac{\widehat{\xi}_{\tau_n}^{\mathrm{pool},\star}(\boldsymbol{\nu}_n, \widehat{\boldsymbol{\omega}}_n)}{\xi_{\tau_n}} = A_n(\widehat{\boldsymbol{\omega}}_n) + B_n(\boldsymbol{\nu}_n) - C_n.$$

Proposition 2.14 gives the random-weight relative-error limit for the pooled-Weissman component. With

$$x_n(\widehat{\boldsymbol{\omega}}_n) = \frac{\widehat{q}_n^\star(1 - p_n \mid \widehat{\boldsymbol{\omega}}_n)}{Q(1 - p_n)} - 1,$$

this gives

$$\frac{\sqrt{k}}{\ell_n} x_n(\widehat{\boldsymbol{\omega}}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Since $\sqrt{k}/\ell_n \to \infty$, $x_n(\widehat{\boldsymbol{\omega}}_n) = O_P(\ell_n/\sqrt{k}) = o_P(1)$, and the relative-to-log Taylor step used in the proof of Theorem 3.2 gives

$$\frac{\sqrt{k}}{\ell_n} A_n(\widehat{\boldsymbol{\omega}}_n) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

The bridge-Hill term is unchanged because $\boldsymbol{\nu}_n$ is deterministic, exactly normalised, and convergent. The proof of (3.7) therefore applies verbatim. The population bridge term C_n is deterministic and is still controlled by (3.8). Slutsky's theorem gives the log limit, and the final exponential Taylor step in Theorem 3.2 gives the relative-error version. □

Thus any source-admissible Daouia–Padoan–Stupfler construction of $\widehat{\boldsymbol{\omega}}_n$ transfers in the pooled-Weissman role, including estimated variance- and AMSE-optimal weights when their source assumptions give $\widehat{\boldsymbol{\omega}}_n \xrightarrow{\mathbb{P}} \boldsymbol{\omega}$. This statement concerns $\boldsymbol{\omega}$ only. It does not optimise $\boldsymbol{\nu}$, and it does not cover a random bridge-Hill convention such as $\boldsymbol{\nu} = \widehat{\boldsymbol{\omega}}_n$.

Corollary 3.6 (Plug-in studentisation and log-scale interval)**.** *Let*

$$\widehat{\xi}_{\tau_n}^{\mathrm{plug}} = \widehat{\xi}_{\tau_n}^{\mathrm{pool},\star}(\boldsymbol{\nu}_n^k, \widehat{\boldsymbol{\omega}}_n), \qquad \boldsymbol{\nu}_n^k = \left(\frac{k_1}{k}, \dots, \frac{k_m}{k}\right)^\top.$$

Under the assumptions of Corollary 3.5, let $\widehat{\boldsymbol{B}}_{\boldsymbol{c}n}^{\mathrm{dist}}$ *and* $\widehat{\boldsymbol{V}}_{\boldsymbol{c}n}^{\mathrm{dist}}$ *be plug-in estimators satisfying*

$$\widehat{\boldsymbol{B}}_{\boldsymbol{c}n}^{\mathrm{dist}} \xrightarrow{\mathbb{P}} \boldsymbol{B}_{\boldsymbol{c}}^{\mathrm{dist}}, \qquad \widehat{\boldsymbol{V}}_{\boldsymbol{c}n}^{\mathrm{dist}} \xrightarrow{\mathbb{P}} \boldsymbol{V}_{\boldsymbol{c}}^{\mathrm{dist}}.$$

These consistency assumptions belong to the selected Daouia–Padoan–Stupfler plug-in route; they are not consequences of the expectile decomposition alone. No expectile-specific bias or variance *estimator is introduced here. Define*

$$\widehat{B}_{\widehat{\boldsymbol{\omega}},n} = \widehat{\boldsymbol{\omega}}_n^\top \widehat{\boldsymbol{B}_{\boldsymbol{c}n}}^{\text{dist}}, \qquad \widehat{V}_{\widehat{\boldsymbol{\omega}},n} = \widehat{\boldsymbol{\omega}}_n^\top \widehat{\boldsymbol{V}_{\boldsymbol{c}n}}^{\text{dist}} \widehat{\boldsymbol{\omega}}_n,$$

and assume $V_{\boldsymbol{\omega}} > 0$. Then

$$T_n^{\text{plug}} = \frac{\frac{\sqrt{k}}{\ell_n}\log\left(\frac{\widehat{\xi}_{\tau_n}^{\text{plug}}}{\xi_{\tau_n}}\right) - \widehat{B}_{\widehat{\boldsymbol{\omega}},n}}{\sqrt{\widehat{V}_{\widehat{\boldsymbol{\omega}},n}}} \xrightarrow{d} \mathcal{N}(0,1).$$

Let $z_{1-\alpha/2}$ be the standard normal quantile. On the high-probability event that the local Weissman estimators are positive and $\widehat{V}_{\widehat{\boldsymbol{\omega}},n} > 0$, define

$$I_{n,1-\alpha}^{\text{plug}} = \left[\widehat{\xi}_{\tau_n}^{\text{plug}} \exp\left\{-\frac{\ell_n}{\sqrt{k}}\left(\widehat{B}_{\widehat{\boldsymbol{\omega}},n} + z_{1-\alpha/2}\sqrt{\widehat{V}_{\widehat{\boldsymbol{\omega}},n}}\right)\right\},\right.$$
$$\left.\widehat{\xi}_{\tau_n}^{\text{plug}} \exp\left\{-\frac{\ell_n}{\sqrt{k}}\left(\widehat{B}_{\widehat{\boldsymbol{\omega}},n} - z_{1-\alpha/2}\sqrt{\widehat{V}_{\widehat{\boldsymbol{\omega}},n}}\right)\right\}\right].$$

Then

$$\mathbb{P}\left\{\xi_{\tau_n} \in I_{n,1-\alpha}^{\text{plug}}\right\} \to 1-\alpha.$$

Proof. Corollary 3.5 gives

$$\frac{\sqrt{k}}{\ell_n}\log\left(\frac{\widehat{\xi}_{\tau_n}^{\text{plug}}}{\xi_{\tau_n}}\right) \xrightarrow{d} \mathcal{N}(B_{\boldsymbol{\omega}}, V_{\boldsymbol{\omega}}).$$

Proposition 2.15, applied with $\boldsymbol{B_c} = \boldsymbol{B_c}^{\text{dist}}$ and $\boldsymbol{V_c} = \boldsymbol{V_c}^{\text{dist}}$, gives

$$\widehat{B}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} B_{\boldsymbol{\omega}}, \qquad \widehat{V}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} V_{\boldsymbol{\omega}}.$$

The additional expectile terms have already been absorbed in Corollary 3.5, so the centring and variance objects remain the pooled-Weissman objects supplied by the source theory. Together with $V_{\boldsymbol{\omega}} > 0$, the displayed convergences give the standard normal limit by Slutsky's theorem. Since $\widehat{V}_{\widehat{\boldsymbol{\omega}},n} \xrightarrow{\mathbb{P}} V_{\boldsymbol{\omega}} > 0$, the event $\widehat{V}_{\widehat{\boldsymbol{\omega}},n} > 0$ has probability tending to one. Eventual upper-tail positivity also gives positivity of the high order statistics, and hence of all local Weissman estimators, with probability tending to one. Solving

$$\left|T_n^{\text{plug}}\right| \le z_{1-\alpha/2}$$

for the unknown ξ_{τ_n} gives the displayed multiplicative interval, so its asymptotic coverage follows from the studentised limit. □

Remark 3.7 (Scope of the inference corollary). The interval in Corollary 3.6 is obtained by inverting the plug-in-centred expectile statistic above. It is therefore a consequence of the transferred expectile limit, rather than a direct restatement of the quantile interval in Daouia, Padoan, et al. (2024). The centering and variance objects remain those of the pooled-Weissman component because, under $\eta_n \to 0$, the additional expectile-bridge terms are lower order on that scale. The corollary is consequently conditional on the selected Daouia–Padoan–Stupfler construction providing consistent plug-in objects for B_{ω} and V_{ω}; it does not make an additional claim about simulations, finite-sample calibration, or optimal bridge-Hill weights.

Chapter 4

Finite-sample study

This chapter reports a finite-sample study for the estimator developed in Chapter 3. The purpose is deliberately narrow. We study the pooled extreme-expectile estimator

$$\widehat{\xi}_{\tau}^{\text{pool},\star}(\boldsymbol{\nu}_n^k, \boldsymbol{\omega}) = \psi(\hat{\gamma}_n(\boldsymbol{\nu}_n^k))\widehat{q}_n^{\star}(1-p \mid \boldsymbol{\omega}), \qquad \boldsymbol{\nu}_n^k = (k_1/k, \ldots, k_m/k)^{\top},$$

where $p = 1-\tau$ and $k = \sum_{j=1}^{m} k_j$. Thus the bridge-Hill weights are fixed at the deterministic threshold-allocation convention, whereas the pooled-Weissman weights $\boldsymbol{\omega}$ vary across the Daouia–Padoan–Stupfler choices studied below. The simulations do not claim optimality of $\boldsymbol{\nu}_n^k$, do not use random bridge-Hill weights, and do not revive the intermediate expectile-pooling route discarded earlier.

4.1 Estimators and implementation

For each machine j, the local Hill estimator and Weissman quantile estimator are computed using the `HTailIndex` and `extQuantile` routines from the `ExtremeRisks` package (Padoan 2026). The simulation code only adds the thesis-specific pooling layer: geometric pooling of the local Weissman estimates, the map ψ, the deterministic bridge-Hill convention $\boldsymbol{\nu}_n^k$, and the summary calculations.

The following estimators are recorded in every scenario:

- *equal*: $\omega_j = 1/m$;
- *DPS variance*: $\omega_j = k_j/k$, the finite-sample independent distributed version of the DPS variance weights;
- *DPS AMSE oracle*: the DPS AMSE formula evaluated with the population second-order parameters of the data-generating model;
- *DPS AMSE plug-in*: the same formula with second-order inputs estimated using `evt0` (Belzile et al. 2026);

- *centralised*: the corresponding single-sample estimator computed from the combined sample, used only as a benchmark.

All distributed estimators use the same bridge-Hill vector $\boldsymbol{\nu}_n^k$. In particular, the plug-in AMSE estimator randomises $\boldsymbol{\omega}$ only.

For each estimator, inference is assessed with the log-scale interval from Corollary 3.6. In a Monte Carlo replication, if $\widehat{B}_\omega$ and $\widehat{V}_\omega$ denote the selected DPS centring and variance objects, the interval is

$$\widehat{\xi}\left[\exp\left\{-\frac{\ell}{\sqrt{k}}(\widehat{B}_\omega + z_{0.975}\sqrt{\widehat{V}_\omega})\right\}, \exp\left\{-\frac{\ell}{\sqrt{k}}(\widehat{B}_\omega - z_{0.975}\sqrt{\widehat{V}_\omega})\right\}\right], \qquad \ell = \log\left(\frac{k}{np}\right).$$

The interval is included as a diagnostic of the first-order plug-in route, not as a claim of finite-sample calibration.

4.2 Design

The study stays within the iid common-marginal setting of the theorem. The data-generating models are Pareto with $\gamma = 0.25$, Pareto with $\gamma = 0.60$, Burr with $\gamma = 0.25$ and $\rho = -1$, Burr with $\gamma = 0.60$ and $\rho = -1$, and the absolute Student distribution with three degrees of freedom. The latter has $\gamma = 1/3$. The true expectile ξ_τ is computed numerically from the population expectile equation

$$\tau\ \mathbb{E}[(X - \xi_\tau)_+] = (1-\tau)\ \mathbb{E}[(\xi_\tau - X)_+],$$

rather than by the asymptotic expectile–quantile bridge.

The final simulation uses 5000 Monte Carlo replications per scenario. The main design has $n = 10000$, $k_j/n_j = 0.05$, $np = 5$, $m \in \{5, 10\}$, and either balanced allocation or a strong imbalanced allocation with machine proportions increasing geometrically from 1 to 10. Two sensitivity blocks then fix $m = 10$ and the strong allocation: one varies $k_j/n_j \in \{0.03, 0.05, 0.10\}$, and the other varies $np \in \{1, 5, 10\}$.

Performance is measured by the Monte Carlo mean of the log relative error, the root mean squared log relative error, empirical coverage of the nominal 95% log interval, average log interval width, invalid-estimate rate, and weight-stability diagnostics.

4.3 Point estimation

Table 4.1 reports the representative main design with $m = 10$, strong sample-size imbalance, $k_j/n_j = 0.05$, and $np = 5$. The distributed DPS variance estimator tracks the centralised benchmark closely across all five models. The log RMSE is essentially unchanged in the Burr heavy case, the Student case, and the Pareto heavy case, and differs only slightly in the lighter-tailed Pareto and Burr settings.

The plug-in AMSE weights do not deliver a systematic improvement over the variance weights in this design. This is not surprising: in the common distributed setting the variance allocation is

Table 4.1: Point-estimation performance in the representative main design ($m = 10$, strong allocation, $k_j/n_j = 0.05$, $np = 5$).

DGP	m	Regime	Estimator	Log bias	Log RMSE	Valid rate
Burr heavy	10	strong	Centralised	0.077	0.226	1.000
Burr heavy	10	strong	DPS AMSE plug-in	0.075	0.223	1.000
Burr heavy	10	strong	DPS variance	0.075	0.223	1.000
Burr light	10	strong	Centralised	-0.039	0.069	1.000
Burr light	10	strong	DPS AMSE plug-in	-0.041	0.070	1.000
Burr light	10	strong	DPS variance	-0.041	0.070	1.000
Pareto heavy	10	strong	Centralised	-0.007	0.204	1.000
Pareto heavy	10	strong	DPS AMSE plug-in	-0.012	0.205	1.000
Pareto heavy	10	strong	DPS variance	-0.012	0.204	1.000
Pareto light	10	strong	Centralised	-0.069	0.089	1.000
Pareto light	10	strong	DPS AMSE plug-in	-0.071	0.091	1.000
Pareto light	10	strong	DPS variance	-0.071	0.090	1.000
Abs. Student-3	10	strong	Centralised	0.110	0.143	1.000
Abs. Student-3	10	strong	DPS AMSE plug-in	0.109	0.142	1.000
Abs. Student-3	10	strong	DPS variance	0.109	0.142	1.000

already the dominant stable choice, and the estimated second-order quantities add finite-sample noise. The conclusion is therefore practical rather than theoretical. The simulation supports the use of DPS variance weights as a stable baseline for $\boldsymbol{\omega}$; it does not suggest opening a new optimisation layer for $\boldsymbol{\nu}$.

4.4 Plug-in interval performance

Table 4.2 shows that the first-order log intervals are not uniformly well calibrated at this sample size. Coverage is close to nominal only in the absolute Student case, where the DPS variance and plug-in AMSE intervals cover about 92% of replications. In the Pareto and Burr cases, coverage is materially below 95%, especially for the lighter-tailed models. The centralised benchmark displays the same qualitative undercoverage, so this is not mainly a distributed-pooling artefact. It is a finite-sample calibration issue for the first-order log approximation and the plug-in centring objects.

The interval results therefore support a cautious interpretation of Corollary 3.6. The corollary gives the asymptotic studentisation when the DPS source route supplies consistent plug-in objects; the finite-sample study shows that the resulting interval can still be noticeably under-calibrated in moderate samples and biased tail models.

Figure 4.3 gives a targeted diagnostic for this undercoverage. It evaluates the same oracle first-order interval against both the exact population expectile and the first-order bridge target $\psi(\gamma)Q(1-p)$. In the light Pareto and Burr designs, much of the coverage loss against the exact expectile disappears when the bridge target is used, and exact-target coverage improves in the larger-n, smaller k/n diagnostic design. This supports the interpretation that the main undercoverage is a plausible finite-sample bridge and first-order calibration effect, not a sign error

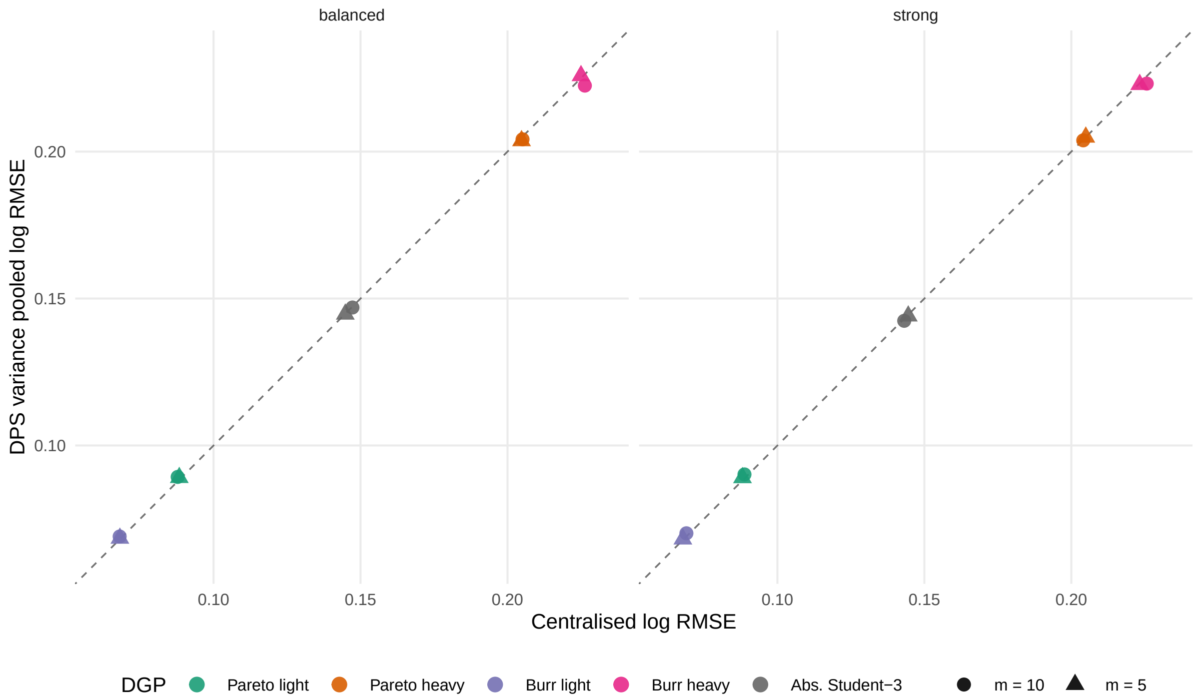

Figure 4.1: Log RMSE of the DPS variance pooled estimator against the centralised benchmark in the main grid with $k_j/n_j = 0.05$ and $np = 5$. The dashed line is equality. Points near the line indicate that the pooled route tracks the centralised benchmark closely.

Table 4.2: Empirical coverage and log-width of nominal 95% intervals in the representative main design.

DGP	m	Regime	Estimator	Coverage	Log width	CI rate
Burr heavy	10	strong	Centralised	0.745	0.498	1.000
Burr heavy	10	strong	DPS AMSE plug-in	0.751	0.498	1.000
Burr heavy	10	strong	DPS variance	0.751	0.498	1.000
Burr light	10	strong	Centralised	0.859	0.208	1.000
Burr light	10	strong	DPS AMSE plug-in	0.591	0.208	1.000
Burr light	10	strong	DPS variance	0.591	0.208	1.000
Pareto heavy	10	strong	Centralised	0.765	0.486	1.000
Pareto heavy	10	strong	DPS AMSE plug-in	0.738	0.491	1.000
Pareto heavy	10	strong	DPS variance	0.738	0.486	1.000
Pareto light	10	strong	Centralised	0.692	0.202	1.000
Pareto light	10	strong	DPS AMSE plug-in	0.671	0.205	1.000
Pareto light	10	strong	DPS variance	0.671	0.202	1.000
Abs. Student-3	10	strong	Centralised	0.691	0.305	1.000
Abs. Student-3	10	strong	DPS AMSE plug-in	0.920	0.305	1.000
Abs. Student-3	10	strong	DPS variance	0.920	0.305	1.000

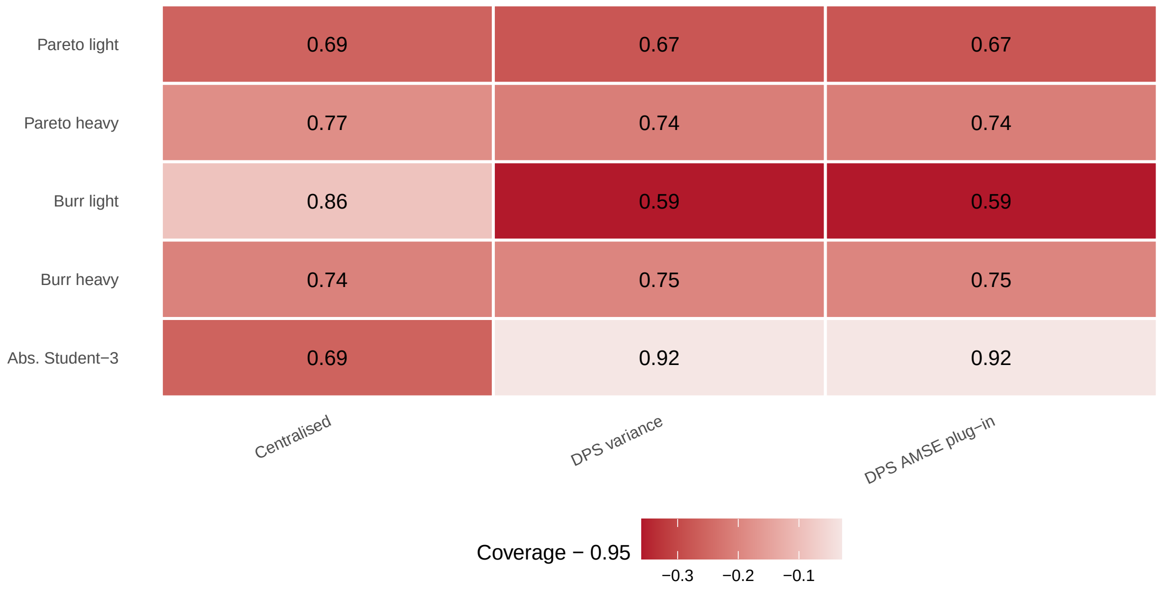

Figure 4.2: Empirical coverage in the representative main design ($m = 10$, strong allocation, $k_j/n_j = 0.05$, $np = 5$). Cell labels give coverage; colour records the deviation from the nominal 95% level.

or a failure of the pooling construction. The diagnostic is not uniform across all heavy-tailed designs, so the conclusion remains cautionary rather than corrective.

4.5 Sensitivity to thresholds and targets

The threshold sensitivity in Table 4.3 shows the usual bias–variance trade-off. For the Burr light model, increasing k_j/n_j reduces log RMSE but worsens coverage sharply, consistent with second-order bias becoming visible. For the absolute Student model, the larger threshold 0.10 increases log RMSE. The Pareto heavy case benefits from a larger threshold in RMSE terms, reflecting the absence of second-order bias in the exact Pareto model. No single threshold is uniformly best across models and metrics.

Table 4.4 varies the target through np. The least extreme target, $np = 10$, often has smaller log RMSE than $np = 1$, as expected from the shorter extrapolation distance. Coverage, however, need not improve when the target is less extreme; in the light Burr and light Pareto cases it falls as np increases. This again points to the interaction between second-order bias, threshold choice, and first-order interval centring.

4.6 Weight stability and limitations

Table 4.5 reports the AMSE weight diagnostics in the representative imbalanced design. The oracle and plug-in AMSE weights are stable in this grid: negative weights are essentially absent, and the average maximum absolute weight is close to the largest deterministic allocation weight. The stability of the weights is one reason the point-estimation performance of DPS variance and plug-in AMSE weights is nearly identical in Table 4.1.

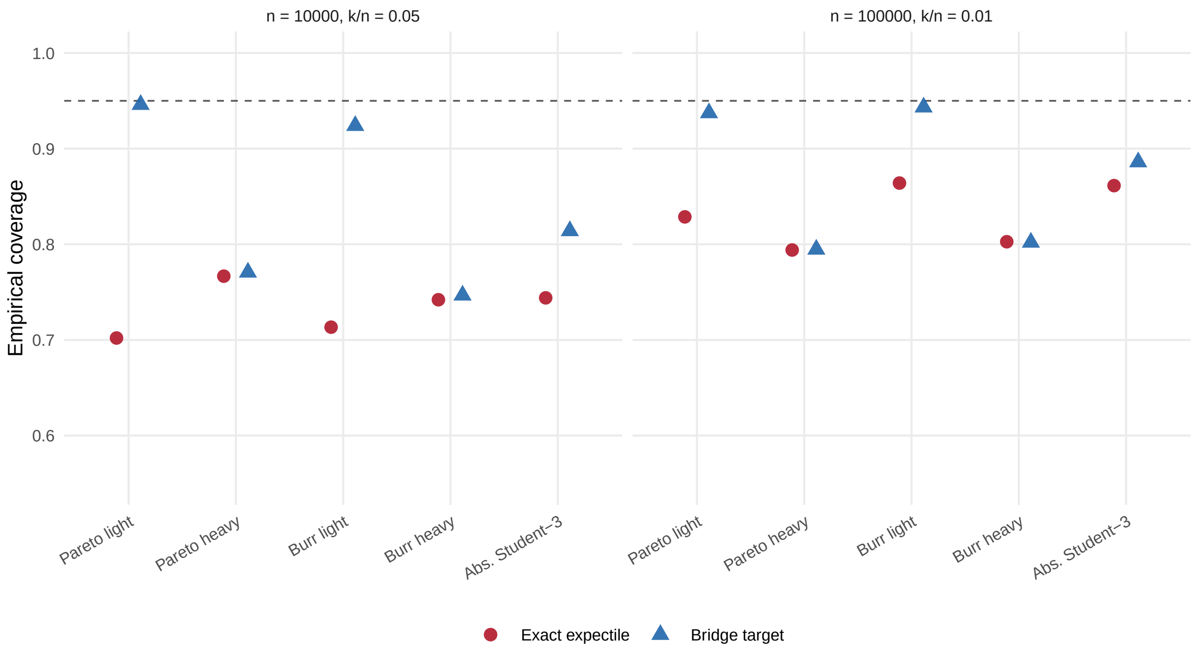

Figure 4.3: Interval diagnostic comparing empirical coverage against the exact population expectile and against the first-order bridge target $\psi(\gamma)Q(1-p)$. The diagnostic uses oracle DPS centring and variance objects and is separate from the final main grid.

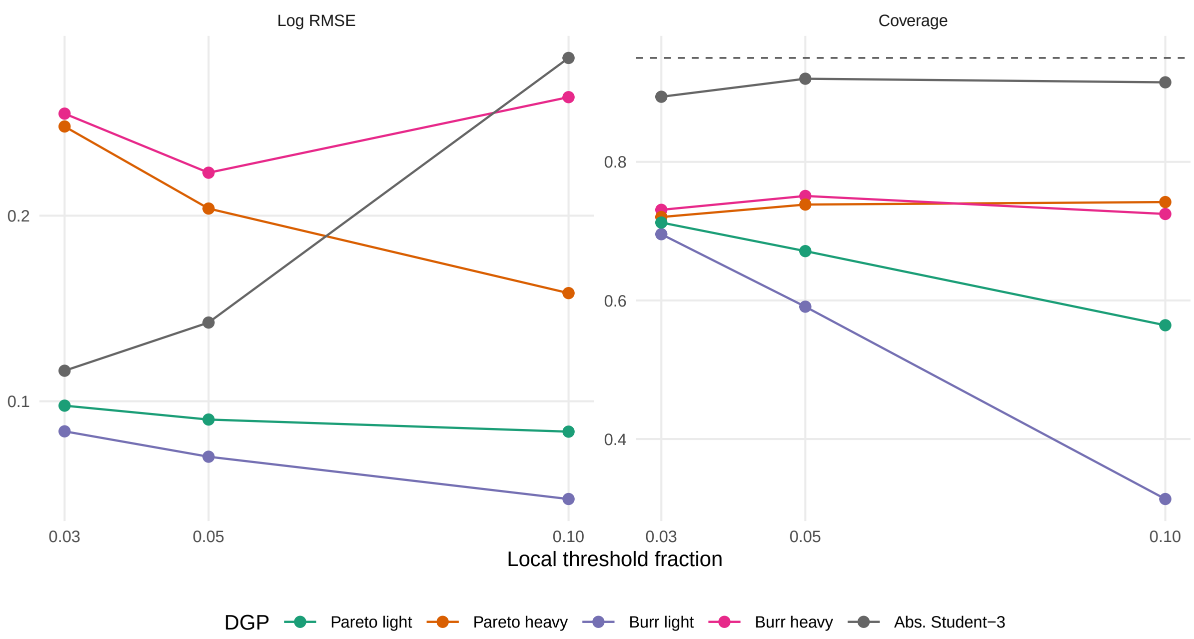

Figure 4.4: Threshold trade-off for the DPS variance estimator with $m = 10$, strong allocation and $np = 5$. The two panels show point-estimation error and interval coverage on the same threshold grid.

Table 4.3: Threshold sensitivity for $m = 10$, strong allocation and $np = 5$.

DGP	k_j/n_j	Estimator	Log RMSE	Coverage
Burr heavy	0.03	DPS AMSE plug-in	0.255	0.731
Burr heavy	0.03	DPS variance	0.255	0.731
Burr heavy	0.05	DPS AMSE plug-in	0.223	0.751
Burr heavy	0.05	DPS variance	0.223	0.751
Burr heavy	0.10	DPS AMSE plug-in	0.264	0.726
Burr heavy	0.10	DPS variance	0.264	0.725
Burr light	0.03	DPS AMSE plug-in	0.084	0.695
Burr light	0.03	DPS variance	0.084	0.696
Burr light	0.05	DPS AMSE plug-in	0.070	0.591
Burr light	0.05	DPS variance	0.070	0.591
Burr light	0.10	DPS AMSE plug-in	0.047	0.314
Burr light	0.10	DPS variance	0.047	0.313
Pareto heavy	0.03	DPS AMSE plug-in	0.249	0.720
Pareto heavy	0.03	DPS variance	0.248	0.720
Pareto heavy	0.05	DPS AMSE plug-in	0.205	0.738
Pareto heavy	0.05	DPS variance	0.204	0.738
Pareto heavy	0.10	DPS AMSE plug-in	0.165	0.742
Pareto heavy	0.10	DPS variance	0.158	0.742
Pareto light	0.03	DPS AMSE plug-in	0.132	0.712
Pareto light	0.03	DPS variance	0.098	0.712
Pareto light	0.05	DPS AMSE plug-in	0.091	0.671
Pareto light	0.05	DPS variance	0.090	0.671
Pareto light	0.10	DPS AMSE plug-in	0.239	0.565
Pareto light	0.10	DPS variance	0.084	0.564
Abs. Student-3	0.03	DPS AMSE plug-in	0.116	0.895
Abs. Student-3	0.03	DPS variance	0.116	0.894
Abs. Student-3	0.05	DPS AMSE plug-in	0.142	0.920
Abs. Student-3	0.05	DPS variance	0.142	0.920
Abs. Student-3	0.10	DPS AMSE plug-in	0.285	0.917
Abs. Student-3	0.10	DPS variance	0.285	0.915

Table 4.4: Target sensitivity for $m = 10$, strong allocation and $k_j/n_j = 0.05$.

DGP	np	Estimator	Log RMSE	Coverage
Burr heavy	1	DPS AMSE plug-in	0.281	0.786
Burr heavy	1	DPS variance	0.281	0.786
Burr heavy	5	DPS AMSE plug-in	0.223	0.751
Burr heavy	5	DPS variance	0.223	0.751
Burr heavy	10	DPS AMSE plug-in	0.207	0.697
Burr heavy	10	DPS variance	0.207	0.697
Burr light	1	DPS AMSE plug-in	0.076	0.775
Burr light	1	DPS variance	0.076	0.776
Burr light	5	DPS AMSE plug-in	0.070	0.591
Burr light	5	DPS variance	0.070	0.591
Burr light	10	DPS AMSE plug-in	0.075	0.429
Burr light	10	DPS variance	0.075	0.429
Pareto heavy	1	DPS AMSE plug-in	0.720	0.790
Pareto heavy	1	DPS variance	0.247	0.790
Pareto heavy	5	DPS AMSE plug-in	0.205	0.738
Pareto heavy	5	DPS variance	0.204	0.738
Pareto heavy	10	DPS AMSE plug-in	0.187	0.716
Pareto heavy	10	DPS variance	0.185	0.716
Pareto light	1	DPS AMSE plug-in	0.165	0.836
Pareto light	1	DPS variance	0.088	0.835
Pareto light	5	DPS AMSE plug-in	0.091	0.671
Pareto light	5	DPS variance	0.090	0.671
Pareto light	10	DPS AMSE plug-in	0.098	0.490
Pareto light	10	DPS variance	0.097	0.490
Abs. Student-3	1	DPS AMSE plug-in	0.225	0.928
Abs. Student-3	1	DPS variance	0.225	0.928
Abs. Student-3	5	DPS AMSE plug-in	0.142	0.920
Abs. Student-3	5	DPS variance	0.142	0.920
Abs. Student-3	10	DPS AMSE plug-in	0.110	0.882
Abs. Student-3	10	DPS variance	0.110	0.883

Table 4.5: AMSE weight diagnostics in the representative main design.

DGP	m	Regime	Estimator	Neg. rate	Avg. max $\lvert\omega\rvert$
Burr heavy	10	strong	DPS AMSE oracle	0.000	0.246
Burr heavy	10	strong	DPS AMSE plug-in	0.000	0.246
Burr light	10	strong	DPS AMSE oracle	0.000	0.246
Burr light	10	strong	DPS AMSE plug-in	0.000	0.246
Pareto heavy	10	strong	DPS AMSE oracle	0.000	0.246
Pareto heavy	10	strong	DPS AMSE plug-in	0.001	0.248
Pareto light	10	strong	DPS AMSE oracle	0.000	0.246
Pareto light	10	strong	DPS AMSE plug-in	0.001	0.248
Abs. Student-3	10	strong	DPS AMSE oracle	0.000	0.241
Abs. Student-3	10	strong	DPS AMSE plug-in	0.000	0.244

The simulation evidence is intentionally limited. It supports the promoted estimator in the iid common-marginal distributed setting and suggests that DPS variance weights are a robust practical choice for $\boldsymbol{\omega}$. It also shows that first-order plug-in intervals may under-cover at realistic sample sizes. The study does not address non-identical margins, time series, multivariate targets, random bridge-Hill weights, or lower-order optimality of $\boldsymbol{\nu}$. Those questions would require additional mathematical work before they could become simulation targets.

Bibliography

Bellini, F., B. Klar, A. Müller, and E. Rosazza Gianin (2014). “Generalized quantiles as risk measures”. In: *Insurance: Mathematics and Economics* 54, pp. 41–48.

Belzile, L., B. G. Manjunath, and F. Caeiro (2026). *evt0: Mean of Order P, Peaks over Random Threshold Hill and High Quantile Estimates*. R package manual, accessed 2026-06-02. URL: https://cran.r-universe.dev/evt0/evt0.pdf.

Daouia, A., S. Girard, and G. Stupfler (2018). “Estimation of tail risk based on extreme expectiles”. In: *Journal of the Royal Statistical Society. Series B* 80.2, pp. 263–292. DOI: 10.1111/rssb.12254.

– (2019). “Extreme M-quantiles as risk measures: From L^1 to L^p optimization”. In: *Bernoulli* 25.1, pp. 264–309. DOI: 10.3150/17-BEJ987.

– (2020). “Tail expectile process and risk assessment”. In: *Bernoulli* 26.1, pp. 531–556. DOI: 10.3150/19-BEJ1137.

Daouia, A., S. A. Padoan, and G. Stupfler (2024). “Optimal weighted pooling for inference about the tail index and extreme quantiles”. In: *Bernoulli* 30.2, pp. 1287–1312. DOI: 10.3150/23-BEJ1632. arXiv: 2111.03173.

Davison, A. C., S. A. Padoan, and G. Stupfler (2023). “Tail risk inference via expectiles in heavy-tailed time series”. In: *Journal of Business & Economic Statistics* 41.3, pp. 876–889. DOI: 10.1080/07350015.2022.2078332.

Ehm, W., T. Gneiting, A. Jordan, and F. Krüger (2016). “Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings”. In: *Journal of the Royal Statistical Society. Series B* 78.3, pp. 505–562.

Gneiting, T. (2011). “Making and evaluating point forecasts”. In: *Journal of the American Statistical Association* 106.494, pp. 746–762.

Haan, L. de and A. Ferreira (2006). *Extreme Value Theory: An Introduction*. Springer Series in Operations Research and Financial Engineering. New York: Springer.

Hill, B. M. (1975). “A simple general approach to inference about the tail of a distribution”. In: *The Annals of Statistics* 3.5, pp. 1163–1174.

Newey, W. K. and J. L. Powell (1987). “Asymmetric least squares estimation and testing”. In: *Econometrica* 55.4, pp. 819–847.

Padoan, S. A. (2026). *ExtremeRisks: Extreme Risk Measures*. R package manual, accessed 2026-06-02. URL: https://cran.r-universe.dev/ExtremeRisks/doc/manual.html.

Weissman, I. (1978). “Estimation of parameters and large quantiles based on the k largest observations”. In: *Journal of the American Statistical Association* 73.364, pp. 812–815.

Ziegel, J. F. (2016). "Coherence and elicitability". In: *Mathematical Finance* 26.4, pp. 901–918. DOI: 10.1111/mafi.12080.